

Quantum-Enhanced Spatio-Temporal Point Process

for Dengue Fever Prediction in Southeast Asia

53K+

STPP EVENTS

8

COUNTRIES

99.46%

SPATIAL CORR.

R2=0.78

FORECAST R2

Quantum Computing

Public Health

Hackathon 2026

Quantum Dengue Team

May 30, 2026

NVIDIA RTX 3090 · 6.3 min GPU

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The Dengue Crisis

A Growing Threat in Southeast Asia

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The Prediction Challenge

Why Classical Models Fail on Dengue Data

Research Questions & Hypothesis

CORE HYPOTHESIS

"Quantum generative models can learn the complex probability distribution of dengue outbreak patterns more efficiently than classical methods — with **exponential advantage potential** as quantum hardware matures."

Research Questions

RQ1

Can quantum generative models learn spatial outbreak probability distributions with high fidelity?

RQ2

Does quantum-augmented data improve spatio-temporal forecasting beyond classical baselines?

RQ3

How does the quantum-inspired pipeline scale vs classical augmentation?

Project Objectives

- ✓ Design QBM + QGAN for disease outbreak augmentation
- ✓ Validate via MMD loss (QBM) and spatial correlation (QGAN)
- ✓ Generate synthetic patterns preserving spatial autocorrelation
- ✓ Benchmark against Transformer-based classical forecaster

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Hybrid Pipeline Architecture

Quantum Generator + Classical Forecaster

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Quantum Born Machine

Learning Spatial Probability Distributions

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Grid QGAN v3

Full Spatial Grid Generation

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Experimental Results

QBM Convergence + Dataset Overview

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Forecasting Performance

Transformer vs Classical Baselines

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Limitations & Roadmap

Honest Assessment

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